

Introduction to Riemannian Manifolds Selected Exercise Solutions

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Chapter 4

4-4 a) A is a $\binom{2}{1}$ tensor field because it is linear over $C^\infty(M)$ both in X and Y . Then, this shows that if ∇^0 is any linear connection on M , then set of all linear connections on M is given by $\{\nabla^0 + A : A \in \mathcal{T}_1^2(M)\}$

b) Assume ∇^0 and ∇^1 define the same geodesics. Let $X, Y \in \mathfrak{X}(M)$. Consider $A(X + Y, X + Y)$. On any solution curve γ to $X + Y$, $A(\dot{\gamma}(t), \dot{\gamma}(t)) = 0$ by the assumption. Since this is true for any solution curve γ with any initial point $p \in M$, we must have $A(X + Y, X + Y) = 0$. Then, a similar argument shows that $A(X, X) = A(Y, Y) = 0$. Then we see that

$$\begin{aligned} 0 &= A(X + Y, X + Y) \\ &= A(X, X) + A(X, Y) + A(Y, X) + A(Y, Y) \\ &= A(X, Y) + A(Y, X) \end{aligned}$$

which proves A is anti-symmetric. Conversely, if we take a geodesic γ in the connection ∇^0 , then by assumption, we have, by taking $X = \dot{\gamma}(t), Y = 0$, we have

$$\begin{aligned} 0 &= A(X, X) \\ &= A(\dot{\gamma}(t), \dot{\gamma}(t)) \\ &= \nabla_{\dot{\gamma}}^0 \dot{\gamma} - \nabla_{\dot{\gamma}}^1 \dot{\gamma} \\ &= -\nabla_{\dot{\gamma}}^1 \dot{\gamma} \end{aligned}$$

which shows that γ is also a connection for the connection ∇^1 .

c) This is a straight computation.

4-5 We can write $X = X^j E_j$ in a neighbourhood of a point $p \in M$. Then, in coordinates, we can figure out the formula for ω_i^k . We compute

$$\begin{aligned} \nabla_X E_i &= X^j \nabla_{E_j} E_i \\ &= X^j \Gamma_{ji}^k E_k \end{aligned}$$

Then we can set $\omega_i^k(X)(q) = \sum_{j=1}^n X^j(q) \Gamma_{ji}^k(q)$ on U .

Chapter 5

Exercise 5-7 a)

$$f(\theta, \varphi) = (R \sin \varphi \cos \theta, R \sin \varphi \sin \theta, R \cos \varphi)$$

To compute the round metric $\tilde{g}$ in spherical coordinates, we pullback the Euclidean metric via f . We have

$$\begin{aligned} f_* \partial_\theta &= -R \sin \varphi \sin \theta \partial_x + R \sin \varphi \cos \theta \partial_y \\ f_* \partial_\varphi &= R \cos \varphi \cos \theta \partial_x + R \cos \varphi \sin \theta \partial_y - R \sin \varphi \partial_z \end{aligned}$$

Then, we compute using the identity $\tilde{g}_{ij} = g(f_* \partial_i, f_* \partial_j)$ to get

$$\tilde{g} = R^2 \sin^2 \varphi d\theta^2 + R^2 d\varphi^2$$

b) To compute the Christoffel symbols we need the inverse of the matrix for $\tilde{g}$.

$$\begin{aligned} (\tilde{g}_{ij}) &= \begin{pmatrix} R^2 \sin^2 \varphi & 0 \\ 0 & R^2 \end{pmatrix} \\ (\tilde{g}_{ij})^{-1} &= \begin{pmatrix} \frac{1}{R^2 \sin^2 \varphi} & 0 \\ 0 & \frac{1}{R^2} \end{pmatrix} \end{aligned}$$

We have the following rule for the christoffel symbols of the Riemannian connection.

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

We get that

$$\begin{aligned} \Gamma_{\theta\theta}^\theta &= 0 \\ \Gamma_{\theta\varphi}^\theta &= \frac{\cos \varphi}{\sin \varphi} \\ \Gamma_{\varphi\varphi}^\theta &= 0 \\ \Gamma_{\varphi\theta}^\varphi &= 0 \\ \Gamma_{\varphi\varphi}^\varphi &= 0 \\ \Gamma_{\varphi\theta}^\theta &= \frac{\cos \varphi}{\sin \varphi} \\ \Gamma_{\theta\varphi}^\varphi &= 0 \\ \Gamma_{\theta\theta}^\varphi &= -\sin \varphi \cos \varphi \end{aligned}$$

c) Each meridian is given by $(\theta(t), \varphi(t)) = (\theta_0, t)$. Then we compute $\dot{\theta} = 0, \dot{\varphi} = 1$. The geodesic equation is

$$\ddot{x}^k + \dot{x}^i \dot{x}^j \Gamma_{ij}^k = 0$$

It is easily seen that the equation is satisfied. Hence, each meridian is a geodesic.

5-1 Skipped because requires the knowledge of total covariant derivatives.

5-2 Homework 2 problem. Will be written here soon.

5-3 a) The geodesics in the hyperboloid model $\mathbb{H}_R^n$ are the intersections of $\mathbb{H}_R^n$ with the 2-planes through the origin. The analog of cos and sin in the hyperboloid model are the cosh and sinh functions. Then, the geodesic curve is given by

$$\gamma(t) = (0, \dots, 0, R \sinh(t/R), R \cosh(t/R))$$

b) Let $p \in \mathbb{H}_R^n$ and $V \in T_p M$. Since $O_+(n, 1)$ acts transitively on $\mathbb{H}_R^n$, we can find a matrix $A \in O_+(n, 1)$ such that $Ap = N := (0, \dots, 0, R)$. Then, we know that the geodesic defined in part a exists for all time t , since the action of A preserves the metric, the pushforward $A_*\gamma$ is the unique geodesic with initial point p and initial velocity vector V .

5-4 a) Let $q \in \mathbb{R}^n$. Take an arbitrary curve with initial point p and endpoint q . Then, parallel transport V_p along this curve to q .

b) Since $\{\partial_\theta, \partial_\varphi\}$ is a local frame for TU on $U \subset S^2$, we have

$$\begin{aligned}\nabla_{\partial_\varphi} \partial_\varphi &= \Gamma_{\varphi\varphi}^\varphi \partial_\varphi + \Gamma_{\varphi\varphi}^\theta \partial_\theta = 0 \\ \nabla_{\partial_\theta} \partial_\varphi &= \Gamma_{\theta\varphi}^\theta \partial_\theta + \Gamma_{\theta\varphi}^\varphi \partial_\varphi = \cot \varphi \cdot \partial_\theta\end{aligned}$$

A meridian is given by $\gamma(t) = (\theta_0, \varphi(t))$. In this case, $\dot{\gamma}(t) = \dot{\varphi}(t)\partial_\varphi$. Then,

$$D_t V(t) = \nabla_{\dot{\gamma}(t)} \partial_\varphi = \dot{\varphi}(t) \nabla_{\partial_\varphi} \partial_\varphi = 0$$

Thus, $V(t)$ is parallel to the meridians. The equator is given by the curve $\beta(t) = (\theta(t), \pi/2)$. Then,

$$D_t V(t) = \nabla_{\dot{\beta}(t)} \partial_\varphi = \dot{\theta}(t) \nabla_{\partial_\theta} \partial_\varphi = \dot{\theta}(t) \cot(\pi/2) \partial_\theta = 0$$

Thus, equator is also a geodesic of S^2 .

c) Assume a parallel extension W to V_p exists in a neighborhood $U \subset S^2$ of p . Then, we can write $W = W^1 \partial_\theta + W^2 \partial_\varphi$ in U . Consider the curve $\gamma(t) = (0, \varphi(t) = t)$. Then, $\dot{\gamma}(t) = \frac{\partial}{\partial \varphi}$. Let's also consider the curve $\beta(t) = (\theta(t) = t, \pi/2 - \epsilon)$ for $\epsilon > 0$ small enough so that $\gamma_V(\epsilon) \in U$. Then, $\dot{\beta}(t) = \frac{\partial}{\partial \theta}$. Since W is a parallel vector field, we must have $\nabla_{\dot{\beta}} W = \nabla_{\dot{\gamma}} W = 0$. We first compute the necessary Christoffel symbols

$$\begin{aligned}\nabla_{\partial_\varphi} \partial_\varphi &= \Gamma_{\varphi\varphi}^\varphi \partial_\varphi + \Gamma_{\varphi\varphi}^\theta \partial_\theta = 0 \\ \nabla_{\partial_\theta} \partial_\theta &= \Gamma_{\theta\theta}^\theta \partial_\theta + \Gamma_{\theta\theta}^\varphi \partial_\varphi = -\sin \varphi \cos \varphi \cdot \partial_\varphi\end{aligned}$$

Now,

$$\begin{aligned}\nabla_{\partial_\varphi} (W^1 \partial_\theta + W^2 \partial_\varphi) &= \partial_\varphi(W^1) \partial_\theta + W^1 \cot \varphi \cdot \partial_\theta + \partial_\varphi(W^2) \partial_\varphi = 0 \\ \nabla_{\partial_\theta} (W^1 \partial_\theta + W^2 \partial_\varphi) &= \partial_\theta(W^1) \partial_\theta - W^1 \sin \varphi \cos \varphi \cdot \partial_\varphi + \partial_\theta(W^2) \partial_\varphi + W^2 \cot \varphi \cdot \partial_\theta = 0\end{aligned}$$

For these equations to hold, the functions W^1, W^2 must satisfy,

$$\partial_\varphi(W^2) = 0 \quad (1)$$

$$\partial_\varphi(W^1) + W^1 \cot \varphi = 0 \quad (2)$$

$$\partial_\theta(W^1) + W^2 \cot \varphi = 0 \quad (3)$$

$$\partial_\theta(W^2) - W^1 \sin \varphi \cos \varphi = 0 \quad (4)$$

By equation (1), we have $\frac{W^2}{\partial_\varphi \partial \theta} = 0$. However, by equation (4), we have

$$\begin{aligned} \frac{\partial^2 W^2}{\partial \varphi \partial \theta} &= \frac{\partial^2 W^2}{\partial \theta \partial \varphi} \\ &= \partial_\varphi W^1 \sin \varphi \cos \varphi + W^1 (\cos^2 \varphi - \sin^2 \varphi) \\ &= -W^1 \cot \varphi \sin \varphi \cos \varphi + W^1 (\cos^2 \varphi - \sin^2 \varphi) \quad \text{by (3)} \\ &= -W^1 \sin^2 \varphi \\ &\neq 0 \end{aligned}$$

Therefore, we see that V_p has no parallel extension in a neighborhood of p .

5-5 Let $X = \text{grad} f$. We are given that $\langle X, X \rangle = 1$. Let γ be a solution curve to X . Then, $\dot{\gamma}(t) = X_{\gamma(t)}$. Then,

$$\frac{d}{dt} \langle X, X \rangle = \langle \nabla_X X, X \rangle + \langle X, \nabla_X X \rangle = 0$$

This is only possible if $X = 0$ or $\nabla_X X = 0$. If it is the former, then γ is a geodesic since $\nabla_{\dot{\gamma}(t)} \dot{\gamma}(t) = \nabla_X X = \nabla_0 0 = 0$. If it is the latter, then we automatically have $\nabla_{\dot{\gamma}(t)} \dot{\gamma}(t) = \nabla_X X = 0$. Hence, we see that the solution curves to X are indeed geodesics.

5-7 Choose a normal neighbourhood and coordinates (U, g) near p . We prove that $\varphi = \psi$ on U first. Let (V, f) be a normal neighbourhood and coordinate chart around $q := \psi(p) = \varphi(p)$ such that $V \subset \psi(U) \cap \varphi(U)$. Then, choose $r \in U$ such that $\varphi(r) \in V$ and $\psi(r) \in V$. Then there is a geodesic from p to r . Say $r = \gamma_V(t)$ for some tangent vector $V \in T_p M$ and $t \in [0, 1]$. Then, in normal coordinates,

$$g(r) = g \circ \gamma_V(t) = tV$$

Since φ and ψ are isometries, they send geodesics to geodesics, hence, using $d\varphi_p = d\psi_p$, we must have

$$\begin{aligned} f \circ \varphi \circ \gamma_V(t) &= t \cdot d\varphi_p(V) \\ &= t \cdot d\psi_p(V) \\ &= f \circ \psi \circ \gamma_t(V) \\ &\iff \\ \varphi(\gamma_V(t)) &= \psi(\gamma_V(t)) \\ &\iff \\ \varphi(r) &= \psi(r) \end{aligned}$$

Thus, we see that $\varphi = \psi$ on U . Now, since M is connected, we can find a smooth curve γ from p to any point on M . Then γ is a compact set, hence we can cover γ with finitely many open sets, on each of which we can apply the previous argument to show $\psi = \varphi$. This then proves that $\varphi = \psi$ on all of M .

Chapter 6

Exercise 6-2 a) Let $\tilde{\gamma} : [0, l] \rightarrow M$ be given by

$$\tilde{\gamma}(t) = \gamma \circ s^{-1}(t)$$

Firstly, $s(t) := \int_a^t |\dot{\gamma}(t)| dt$ is an increasing function and therefore has an inverse. Secondly, we see that

$$\begin{aligned} \tilde{\gamma}'(t) &= \dot{\gamma}(s^{-1}(t)) \cdot (s^{-1})'(t) \\ &= \dot{\gamma}(s^{-1}(t)) \cdot \frac{1}{\dot{s}(s^{-1}(t))} \\ &= \frac{\dot{\gamma}(s^{-1}(t))}{|\dot{\gamma}(s^{-1}(t))|} \end{aligned}$$

which is obviously unit-speed. Hence, $\tilde{\gamma}$ is a unit-speed parametrization of the curve γ . It is also an orientation preserving curve because $\tilde{\gamma}'(t)$ and $\dot{\gamma}(t)$ have the same sign.